

grity of Galaxy Sports Edge (GSE). An institutional sports intelligence platform cannot substitute statistical hygiene for an active live ingestion engine, nor can it mask customer-facing prediction inversion behind historical backtest performance.

$$\text{Composite Score} = \sum_{i=1}^8 w_i \cdot S_i$$

Each dimension $S_i \in [0, 100]$ is evaluated against explicit empirical standards. A score of 0 denotes an immediate fatal blocker or complete operational failure; 50 denotes honest parity with efficient closing market benchmarks; and 100 represents a fully automated, mathematically verified, and externally auditable production state.

- **Customer-Facing Calibration ($w_1 = 0.20$):** Measures whether displayed numbers represent strictly monotonic, well-calibrated probabilities. Publishing inverted or anti-predictive confidence metrics directly degrades user decision quality and invalidates platform credibility.
- **CLV and Market Honesty ($w_2 = 0.15$):** Evaluates the mathematical consistency of Closing Line Value (CLV) accounting, the transparency of matched versus unmatched closes in the denominator, and the absence of fabricated track records.
- **NFL Physical Model ($w_3 = 0.15$):** Evaluates the core margin and total estimation pipeline against efficient market closing prices across seasonal phases (early-season prior reliance versus mid-season regression), quarterback transitions, and environmental dynamics.
- **NCAAF Pipeline ($w_4 = 0.10$):** Assesses pipeline coverage, schema integrity, and predictive validity for college football slates, heavily penalizing unmonitored failures and missed slates.
- **MLB Pipeline and Copy Honesty ($w_5 = 0.10$):** Measures market pricing validity across run lines and moneylines, market depth scoring accuracy, and the elimination of tautological or misleading product copy.
- **Fantasy and Player Props ($w_6 = 0.10$):** Evaluates the physical modeling of count distributions, proper treatment of variance overdispersion, environmental factor decoupling, and live quote ingestion integrity.
- **Ops and System Reliability ($w_7 = 0.10$):** Evaluates ingestion pipeline uptime, serverless execution limits, database indexing, memory stability, and truth surface availability.
- **Statistical Hygiene and Governance ($w_8 = 0.10$):** Measures adherence to out-of-sample testing, rejection of noise-mining features, rigorous feature-kill governance, and enforcement of non-AI positioning rules.

Quantitative Audit Breakdown

Dimension	Weight (w_i)	Score (S_i)	Weighted Contribution	Primary Defect or Verification Anchor
(i) Customer-Facing Calibration	0.20	12	2.40	Confidence metric is strongly anti-predictive ($n=2,385$, $z=-10.7$, Brier score 0.3617). Card displays an

Dimension	Weight (w _i)	Score (S _i)	Weighted Contribution	Primary Defect or Verification Anchor
				inverted score.
(ii) CLV / Market Honesty	0.15	28	4.20	As-written CLV stands at 23.2% [0.212, 0.253] due to MATCHED_CLOSE denominator contamination; decided-only CLV is 40.8% [0.377, 0.441] against a 52.4% floor; 21-day archive gap.
(iii) NFL Physical Model	0.15	42	6.30	Market close CRPS (7.109) outperforms all internal structural models (DAVE k=8 yields 7.500); scale parameter divergence (36.61 pooled OLS vs. 45.42 Hermes 2025); live settled fixtures n \approx 70.
(iv) NCAAF Pipeline	0.10	0	0.00	Slate of 2026-09-19 completely missed; zero historical training corpus wired; literature reflects no empirical arXiv calibration papers.
(v) MLB Pipeline & Copy	0.10	25	2.50	Moneyline ECE is healthy (0.0501), but SPREAD 10+ is anti-predictive, run-line consensusPct=1.0 is a structural tautology, and MLB SPEAK 40%

Dimension	Weight (w _i)	Score (S _i)	Weighted Contribution	Primary Defect or Verification Anchor
				is inverted (n=90).
(vi) Fantasy & Player Props	0.10	34	3.40	Red-zone touchdown variance-to-mean ratio (1.82–1.92) confirms Negative Binomial overdispersion and rejects Poisson; quotes are landing, but wind adjustments are not properly isolated from game totals.
(vii) Ops & Reliability	0.10	20	2.00	Ingestion watchdog in RED alarm since 2026-09-13 16:53 UTC; serverless out-of-memory crash on 1.64M rows; public truth surface dark.
(viii) Statistical Hygiene	0.10	82	8.20	12/12 Move-37 lab feature kills upheld; W5–W8 noise rejected; zero-sum failure documented; non-AI product copy discipline enforced.
Total Composite Score	1.00	—	29.00 / 100	System Unfit for Production Deployment

Arithmetic Synthesis

$\text{Composite Score} = 2.40 + 4.20 + 6.30 + 0.00 + 2.50 + 3.40 + 2.00 + 8.20 = 29.00$

The empirical score today is **29.00 out of 100**. While statistical hygiene remains commendable at 82/100 due to rigorous laboratory pruning, production reliability (20/100) and customer-facing calibration (12/100) severely undermine the platform's core utility. The previous internal readiness score of 38 failed to penalize the complete outage of the truth surface following the serverless out-of-memory (OOM) failure on September 13, and it overlooked the finding that client pick cards display a confidence metric that is inversely correlated with realized win rates (z

= -10.7). A score of 29.00 defines a research laboratory with valid statistical instincts whose production deployment remains non-functional and uncalibrated.

B. State of the Architecture (2026-09-19 23:00 CDT)

The operational and analytical infrastructure is split across two disjoint environments: the production engineering environment (Tree A) and the research mill desk (Tree B).

In Tree A ([span_31](start_span)[span_31](end_span)Beexly/Sports), 7.7 million raw odds snapshot rows reside in production databases, where an uncapped ingestion query caused serverless memory exhaustion, triggering a sustained watchdog alarm and disabling calibration reporting. Customer cards currently display an inverted confidence metric, while PR #866, PR #867, and PR #868 remain unmerged.

In Tree B (Research Mill Desk), 14,251 decided NFL fixtures (1999–2025) are modeled to benchmark continuous ranked probability scores, where closing lines outperform structural models, scale constants diverge between historical and single-season folds, and candidate features W5–W8 have been killed.

The current architectural state is detailed below across three formal evidential categories: Observation, Inference, and Speculation.

Observation

Tree A: Production Engine (Beexly/Sports)

- The odds ingestion watchdog has remained continuously in a RED alarm state since 2026-09-13 16:53 UTC. The odds_lin[span_7](start_span)[span_7](end_span)e[span_35](start_span)[span_35](end_span)_snapshots table accumulated 7.7 million rows without pagination boundaries, and an uncapped IN query attempted to materialize 1.64 million snapshot rows into a serverless instance, resulting in immediate memory exhaustion.
- The calibration cron has been offline for six days, preventing truth surface generation and halting live performance reporting.
- PR #868 (serverless query batching and memory cap), PR #867 (EPA path configuration and founder decision register), and PR #866 (archive freshness monitoring) remain unmerged pending verified continuous integration builds.
- Live customer cards display an uncalibrated confidence figure. Across $n = 2$, [span_39](start_span)[span_39](end_span)385 settled decisions, this metric is strongly anti-predictive, generating a Brier score of 0.3617 with an associated z-score of -10.7.
- In an empirical ordering duel across $n = 689$ picks, sorting by marketFairProb achieves a Brier score of 0.2189, outperforming internal rankingP (0.2462) and internal confidence (0.2741).
- Reported Closing Line Value win rate is recorded at 23.2% (95% CI: [0.212, 0.253]) when MATCHED_CLOSE records are retained in the denominator (686/1,587). Restricting evaluation to decided lines yields 40.8% [0.377, 0.441], falling well below the 52.4% break-even floor.
- A data gap exists in the odds archive between 2026-08-23 and 2026-09-12. Database records show fixture triplication, unfinalized records marked PENDING dating to January, and a last-write timestamp on gate_decisions of 2026-06-11.

Tree B: Research Mill Desk

- Across 14,251 decided NFL fixtures (1999–2025), the closing market line achieves a Continuous Ranked Probability Score (CRPS) of 7.109, outperforming all structural models. DAVE ($k=8$) yields 7.500, pooled EPA yields 7.573, and core Elo yields 7.576.
- The pooled ordinary least squares (OLS) scale factor across the 14,251 fixture corpus is 36.61, whereas the Hermes 2025 single-season fold yields 45.42 (DAVE Brier 0.251 vs. market 0.204).
- Across the 2025 season ($n = 272$), DAVE CRPS of 7.629 underperformed both the closing line (6.927) and Elo (7.563). DAVE outperformed Elo strictly during Weeks 1–4 (+0.326), while underperforming across Weeks 5–18 (-0.187), in games with starting quarterback changes (-0.171), and in blowouts (-0.442).
- The quarterback change residual variance ratio (τ^2) was measured at 1.379 (78/114), failing the pre-registered significance kill threshold of 1.50.
- Hawkes self-exciting process modeling yielded a branching ratio of $\hat{n} = 0$, exhibiting Poisson behavior with a kickoff inhibition parameter of 0.448.
- Empirical environmental checks confirm that wind impacts passing yardage by -3.007 yds/mph , rain increases incomplete pass rates (30.1% vs. 27.6%), and red-zone touchdown variance-to-mean ratios measure 1.82–1.92, which confirms Negative Binomial overdispersion and rejects Poisson assumptions.
- Candidate features W5–W8 (Wasserstein distance, Detrended Fluctuation Analysis, Intrinsic Dimension, and Permutation Entropy) failed pre-registered empirical criteria and were killed.

Inference

- The divergence between the historical pooled OLS scale (36.61) and the modern Hermes fold (45.42) reflects structural shifts in league scoring dynamics, offensive efficiency, and rule adjustments over 26 years. Taking an unweighted linear average of these values (41.01) would miscalibrate the distribution variance (σ), distorting moneyline fair values derived from predicted point spreads.
- The production system's CLV collapse to 23.2% is an operational artifact of treating unmatched line snapshots as failed line-beats. When games lack an automated match against the final closing snapshot, the pipeline records them as non-beats, deflating measured CLV. However, even the true decided-only figure (40.8%) reflects severe market underperformance against competitive sportsbooks.
- The inverse relationship between UX confidence and realized outcomes ($z = -10.7$) indicates that the current ranking function overweights factors that commercial bookmakers deliberately exploit, such as public betting favorites and unadjusted brand biases.

Speculation

- The archive gap from August 23 to September 12 suggests that unmonitored schema mutations or token expirations occurred in the ingest workers during early college football deployment, going undetected due to the lack of heartbeat monitors.
- If the EPA auto-activation scheduled for 2026-10-08 triggers with repository default

constants (scale 0.12, home-field advantage 0.025) rather than the empirically measured values (45.42 and 2.10), the production model will output win probabilities compressed tightly around 50%, producing nonsensical recommendations.

C. Week 3 Launch Gate (2026-09-24)

Binary Gate Verdict

DO NOT OPEN THE CUSTOMER BOARD ON THURSDAY, SEPTEMBER 24, 2026.

Opening the board under current conditions would require exposing an inverted confidence metric to paying customers, relying on an unverified CLV calculation, and deploying models with unmonitored infrastructure. College football operations have already failed for the current week. Re-establishing customer trust requires adhering strictly to verified operational gates rather than marketing deadlines.

Minimum True Checklist to Reverse Verdict

The customer board may transition to **OPEN** status for NFL Week 3 if and only if every condition in the following checklist is resolved, verified by automated end-to-end integration tests, and signed off in the operational repository:

- **Infrastructure and Recovery:**
 - PR #868 is merged to main following a verified green CI build, resolving serverless OOM issues.
 - The odds ingestion query is restricted using cursor-based pagination and a maximum limit of 50,000 rows.
 - Serverless execution memory allocation is increased to 1024MB, with cold-start latency verified under 400ms.
 - The monitoring watchdog confirms a continuous GREEN state for at least 48 consecutive hours.
 - The truth surface calibration cron is re-established and actively writing snapshots.
- **Customer-Facing Calibration Repair:**
 - The legacy inverted confidence display is completely purged from client-side pick cards.
 - Pick card sorting is driven strictly by marketFairProb or devigged consensus odds.
 - Public cards display uncalibrated metrics exclusively as non-probabilistic integer indices.
 - Displayed true probabilities are calibrated via Venn-Abers bounds or isotonic binning.
 - Ordering duel validation confirms a Brier score ≤ 0.2189 on the current active slate.
- **Market Integrity and Reporting:**
 - CLV reporting is updated to display both metrics side-by-side: Total Tracked (23.2% [0.212, 0.253]) and Decided Matches (40.8% [0.377, 0.441]).
 - Explanatory copy is published detailing the mathematical mechanism behind the MATCHED_CLOSE calculation.
 - A public notice formally documenting the 2026-08-23 to 2026-09-12 data archive gap is deployed.

- All marketing copy claiming "AI-powered edge" is purged across all public-facing assets.
- **Baseball and Props Sanitization:** * The consensusPct = 1.0 artifact is purged from MLB run lines, setting consensusScore = 0.
 - MLB market depth scoring is disabled until minimum bookmaker depth criteria are verified.
 - NFL PASS recommendations are suppressed until historical sample size reaches n \ge 100 settled games.
 - Red-zone touchdown prop distributions are parameterized using Negative Binomial models with variance-to-mean ratio \ge 1.82.
- **College Football Ring-Fencing:**
 - NCAAAF customer boards remain completely offline for the upcoming slate.
 - A clear public notice is posted: "NCAAF board offline pending ingestion verification".
 - No college football projections are published without an auditable closing line archive.

D. Dated Through-Path: Road to 100

Transitioning GSE from its current state (29) to an institutional standard (100) requires a disciplined sequence of implementations, kill lines, and methodological integrations. Each step maps verified research papers from Jev's arXiv corpus directly to concrete operational components.

Progression Roadmap

Phase 0: System Stabilization & Remediation (Score: 29 -> 45)

Window: 2026-09-19 to 2026-09-21

- Merge PR #868, PR #867, PR #866.
- Cap odds ingest queries at 50k rows.
- Unpublish inverted confidence score from UI cards.
- Isolate archive hole (2026-08-23 to 2026-09-12) via read-replica audit.

Phase 1: Week 3 Gate & Market

Baseline[span_17](start_span)[span_17](end_span)e (Score: 45 -> 70)

Window: 2026-09-22 to 2026-09-24

- Deploy Shin de-biasing consensus model (Kaunitz et al., arXiv:1710.02824).
- Apply Jackknife+ UQ around closing lines (Barber et al., arXiv:1905.02928).
- Publish dual CLV reporting (23.2% system / 40.8% decided).
- Suppress uncalibrated NCAAAF boards.

Phase 2: Engine v5.3.0 & 14k Fixture Calibration

([span_8](start_span)[span_8](end_span)Score: 70 -> 80)

Window: 2026-09-25 to 2026-10-08

- Enforce scale factor 45.42 and HFA 2.10; sign MODEL_VERSION 5.3.0.

- Implement cross-conformal efficiency (Gasparin & Ramdas, arXiv:2503.01495).
- Wire non-exchangeable conformal quantiles (Barber et al., arXiv:2202.13415).
- Calibrate binary cover heads via Venn-Abers (Vovk & Petej, arXiv:1211.0025).

Phase 3: Multi-Sport Integration & Props Tails (Score: 80 -> 85)
Window: 2026-10-09 to 2026-11-15

- Ingest cfbfastR and CollegeFootballData historical corpus (2014-2025).
- Calibrate CFB spreads via additive quantiles (Sides & Harvill, arXiv:2212.08116).
- Parameterize props via Negative Binomial qgam (Fasiolo et al., arXiv:1707.03307).
- Decouple surface wind adjustments (-3.007 yds/mph) from game totals.

Phase 4: Institutional Auditing Standard (Score: 85 -> 100)
Window: 2026-11-16 to 2027-01-15

- Enforce Conformal Risk Control on drawdowns (Angelopoulos et al., arXiv:2208.02814).
- Correct historical selection bias in profit claims (Clegg & Cartlidge, arXiv:2306.01740).
- Launch public cryptographic verification suite (/performance/audit).
- Complete third-party verification against Pinnacle, Circa, and Sumer benchmarks.

Phase 0: System Stabilization and Remediation (Score 29 to 45)

- **Window:** Saturday 2026-09-19 23:30 CDT to Monday 2026-09-21 18:00 CDT.
- **Step 0.1 (Infrastructure):** Verify CI suite and merge PR #868 to terminate serverless OOM exceptions. Enforce a 50,000-row batching window with cursor pagination on odds_line_snapshots.
 - *Evidence:* Serverless memory profile remains below 450MB over a 24-hour cycle.
 - *Kill Line:* If heap memory consumption exceeds 750MB during scheduled cron execution, terminate ingestion immediately.
 - *Owner:* Ops Lead.
 - *Failure Mode:* Ingestion worker disabled; system holds previous cached snapshot.
- **Step 0.2 (Customer Board Honesty):** Unpublish internal confidence figures from client-side cards. Render marketFairProb derived via Shin devigging as the primary sorting index. Purge consensusPct = 1.0 on MLB run lines.
 - *Evidence:* Production pick cards render valid sorted market bounds without raw probability percentages.
 - *Kill Line:* If UI renders raw model probability before calibration is restored, trigger deployment rollback.
 - *Owner:* Engine Lead.
 - *Failure Mode:* Public cards revert to showing market consensus odds without

internal model projections.

- **Step 0.3 (Truth Surface Diagnostics):** Execute database audit via replica read queries to isolate the August 23–September 12 archive gap and resolve pending fixture triplications.
 - *Evidence:* Diagnostic log outputting exact counts of orphaned, duplicated, and unclosed fixtures.
 - *Kill Line:* If data corruption extends to historical grading tables, quarantine the affected sports.
 - *Owner:* Data Lead.
 - *Failure Mode:* Affected sports locked from public display until records are cleaned.

Phase 1: Week 3 Gate and Market Baseline (Score 45 to 70)

- **Window:** Tuesday 2026-09-22 00:00 CDT to Thursday 2026-09-24 17:00 CDT.
- **Step 1.1 (Market-First Pricing):** Ingest multi-book odds feeds and compute consensus fair lines using the Power and Shin de-biasing methods mapped from Kaunitz et al. (arXiv:1710.02824) and Szalkowski & Nelson (arXiv:1211.4000).
 - *Evidence:* Multi-book vig removal achieves a Brier score ≤ 0.204 on historical closing lines.
 - *Kill Line:* If multi-book fair price fails to beat single-book closing lines across 100 historical test games, fall back to Pinnacle closing line.
 - *Owner:* Engine Lead.
 - *Failure Mode:* Devigging pipeline falls back to Pinnacle midpoint lines.
- **Step 1.2 (Uncertainty Quantification Core):** Deploy Jackknife+ predictive inference mapped from Barber, Candès, Ramdas, & Tibshirani (arXiv:1905.02928) across the 14,251 fixture historical corpus to establish valid confidence intervals around market spreads.
 - *Evidence:* Marginal coverage achieves $1 - \alpha \geq 0.90$ across historical validation folds ($n \geq 150$).
 - *Kill Line:* If out-of-sample coverage drops below 0.85 or interval widths exceed Gaussian-close baselines by $> 15\%$, kill Jackknife+ and default to standard split conformal.
 - *Owner:* Engine Lead.
 - *Failure Mode:* Conformal bands fall back to static Gaussian close intervals ($\pm 1.645 \sigma$).
- **Step 1.3 (Dual CLV Publication):** Relaunch the public accountability tracking page displaying both total tracked CLV (23.2%) and decided CLV (40.8%). Include clear documentation detailing the impact of the historical archive gap.
 - *Evidence:* Public web page verified live by third-party curl requests without authentication walls.
 - *Kill Line:* If marketing attempts to suppress the 23.2% metric, block deployment of the tracking dashboard.
 - *Owner:* Founder.
 - *Failure Mode:* The performance page remains in restricted internal preview.

Phase 2: Engine v5.3.0 and Calibration Engine (Score 70 to 80)

- **Window:** Friday 2026-09-25 08:00 CDT to Thursday 2026-10-08 23:59 CDT.

- **Step 2.1 (EPA Path Parameterization):** Formally sign off on production constants for the EPA auto-activation scheduled for 2026-10-08. Enforce an empirical scale factor of 45.42 and home-field advantage of 2.10 points, completely overriding default repository values (0.12 and 0.025).
 - *Evidence:* Production environment variables register SCALE_CONSTANT=45.42 and HFA_POINTS=2.10.
 - *Kill Line:* If auto-activation executes with scale < 30.0, hard-fail the scoring pipeline.
 - *Owner:* Founder.
 - *Failure Mode:* System blocks EPA integration and holds baseline Elo ratings.
- **Step 2.2 (Cross-Conformal Efficiency):** Implement efficient cross-conformal prediction as formulated by Gasparin & Ramdas (arXiv:2503.01495) to utilize full window observations without burning calibration splits.
 - *Evidence:* Prediction interval width narrows by $\geq 8\%$ relative to standard split conformal while preserving 90% coverage.
 - *Kill Line:* If execution latency exceeds 250ms per slate, revert to pre-computed static conformal tables.
 - *Owner:* Engine Lead.
 - *Failure Mode:* Model defaults to pre-calculated split-conformal quantile cutoffs.
- **Step 2.3 (Non-Exchangeable Temporal Conformal):** Integrate non-exchangeable conformal methods from Barber et al. (arXiv:2202.13415) and Oliveira et al. (arXiv:2203.15885) to handle distribution shifts across successive NFL weeks.
 - *Evidence:* Coverage remains within [0.88, 0.92] across seasonal transitions (Weeks 1–4 to Weeks 5–18).
 - *Kill Line:* If early-season interval width expands by $> 50\%$ relative to mid-season bands, kill adaptive weighting and hold static historical quantiles.
 - *Owner:* Engine Lead.
 - *Failure Mode:* Weekly adaptation is disabled; intervals are calibrated on pooled historical residuals.
- **Step 2.4 (Venn-Abers Binary Calibration):** Deploy Venn-Abers predictors from Vovk & Petej (arXiv:1211.0025) and Petej & Vovk (arXiv:2605.06646) on binary moneyline and spread cover events.
 - *Evidence:* Out-of-sample Brier score improves over uncalibrated rankingP (0.2462) by at least 0.005, approaching or beating the 0.2189 market baseline.
 - *Kill Line:* If Venn-Abers Brier score fails to beat 0.2189 by ≥ 0.005 , do not use it for primary ordering; retain devigged market probability as primary sort.
 - *Owner:* Engine Lead.
 - *Failure Mode:* Probability display reverts to market fair probability.

Phase 3: Multi-Sport Integration and Props Tail Modeling (Score 80 to 85)

- **Window:** Friday 2026-10-09 08:00 CDT to Sunday 2026-11-15 23:59 CDT.
- **Step 3.1 (NCAAF Pipeline Ingestion):** Ingest play-by-play and game logs from cfbfastR and historical odds from collegefootballdata.com (2014–2025). Calibrate college point spreads via additive quantile modeling using methods from Sides & Harvill (arXiv:2212.08116).
 - *Evidence:* Out-of-sample CRPS beats basic Elo baseline across two complete

- testing seasons ($n \geq 1,200$).
 - *Kill Line*: If college closing line beat-rate falls below 48.0%, keep NCAAAF off the public board.
 - *Owner*: Data Lead.
 - *Failure Mode*: NCAAAF product remains dark for customer consumption.
- **Step 3.2 (Player Props Tail Calibration)**: Implement Negative Binomial parameterization for touchdown props (VMR 1.82–1.92) using Fast Additive Quantile Regression (qgam) from Fasiolo, Wood, & Zaffran (arXiv:1707.03307).
 - *Evidence*: Tail probability Brier score improves by ≥ 0.015 over standard Poisson models.
 - *Kill Line*: If fitted dispersion parameter ϕ collapses toward Poisson assumptions ($\text{VMR} \rightarrow 1.0$), discard the tail adjustments.
 - *Owner*: Engine Lead.
 - *Failure Mode*: Touchdown props revert to standard Poisson pricing without premium tail claims.
- **Step 3.3 (Trench Matchup and Weather Integration)**: Wire opponent-adjusted pass protection and pass rush win rates using the formulation from Pipping-Gamón et al. (arXiv:2604.01491). Couple surface wind vectors exclusively to passing prop yards and total distributions.
 - *Evidence*: Wind lookup correctly shifts passing yardage quantiles by -3.007 yds/mpH without double-counting game totals.
 - *Kill Line*: If trench metrics fail to improve margin spread CRPS by at least 0.005, drop them from the model.
 - *Owner*: Data Lead.
 - *Failure Mode*: Trench adjustments are zeroed out in team rating calculations.

Phase 4: Institutional Auditing Standard (Score 85 to 100)

- **Window**: Monday 2026-11-16 08:00 CDT to Friday 2027-01-15 18:00 CDT.
- **Step 4.1 (Conformal Risk Control)**: Deploy Conformal Risk Control as developed by Angelopoulos, Bates, Fisch, & Lei (arXiv:2208.02814) to formally bound expected CLV drawdown and tail loss exposure.
 - *Evidence*: Mathematical proof and test suite establishing finite-sample risk bounds on customer portfolio drawdown.
 - *Kill Line*: If defined risk metric violates monotonicity with respect to set size, discard and report standard coverage sets.
 - *Owner*: Engine Lead.
 - *Failure Mode*: Risk control falls back to marginal coverage intervals without loss-bounding guarantees.
- **Step 4.2 (Selection Bias Correction Audit)**: Apply Clegg & Cartlidge's correction framework (arXiv:2306.01740) across the entire settled database to identify and neutralize selection-bias artifacts in reported profits.
 - *Evidence*: Independent audit verifies that simulated betting strategies maintain positive expected value post-correction.
 - *Kill Line*: If positive returns disappear under selection bias correction, restrict platform claims to pure probability calibration.
 - *Owner*: Founder.
 - *Failure Mode*: Remove all return-on-investment copy; position platform purely as an analytical odds-comparison tool.
- **Step 4.3 (Public Cryptographic Verification Suite)**: Launch a public verification

repository allowing external researchers to recompute every Brier score, ECE curve, and CLV percentage directly from immutable hash-chained database receipts.

- *Evidence*: Verification suite runs cleanly in independent CI environments (GitHub Actions) without proprietary data dependencies.
- *Kill Line*: If public verification fails to replicate published metrics to within 10^{-6} precision, fail the release.
- *Owner*: Ops Lead.
- *Failure Mode*: Verification badge revoked until recalculation matches to floating-point precision.

E. Complete Item Inventory

Every locked measurement, production parameter, and Jev arXiv paper is categorized below with an explicit operational disposition. No items are omitted.

Production Measurements and Architectural Items

Item / Signal / Parameter	Current Value / State	Target / Kill Threshold	Operational Tag	Rationale and Disposition
Closing Line Margin Baseline	CRPS = 7.109 (n=1,871)	Benchmark anchor	keep	Primary benchmark for point margins. Structural models must not override it without measured gains.
DAVE (k=8) Margin Model	CRPS = 7.500 (n=1,871)	Beat Close (7.109)	measure	Retained as Rung-2 input for Weeks 1–4 (+0.326 gain). Barred from serving as primary mean estimator μ .
Pooled EPA Margin Model	CRPS = 7.573	Beat Close (7.109)	measure	Subordinated to market close; provides team strength context.
Core Elo Baseline	CRPS = 7.576	Beat DAVE (7.500)	keep	Provides stable ratings baseline. Outperformed by DAVE early, but superior in Weeks 5–18.
Fay-Herriot Year Shrinkage	$\hat{k} = 7.486$ ($\Delta 0.011$ vs. $k=8$)	$\Delta \text{CRPS} \geq 0.01$	keep	Exceeds the 0.01 improvement threshold.

Item / Signal / Parameter	Current Value / State	Target / Kill Threshold	Operational Tag	Rationale and Disposition
				Retained for season-to-season rating transitions.
Mondrian-k Calibration	$\Delta \text{CRPS} = 0.0092$	$\Delta \text{CRPS} \geq 0.01$	kill	Failed pre-registered improvement threshold of 0.01. Production k remains fixed at 8.
OLS Pool Scale (Tree B)	36.61	Single-universe consistency	measure	Reflects historical 14k fixture corpus. Barred from being averaged with production scale (45.42).
Hermes Fold Scale (Tree A)	45.42	Accurate variance calibration	keep	Accurately models the 2025 modern scoring environment. Must be explicitly signed into v5.3.0 constants.
Repository Default Scale	0.12	Match empirical scale (45.42)	kill	Represents a 3.8x parameter error relative to empirical reality. Deployments containing this value must be blocked.
QB-Change Variance Ratio	$\tau^2 = 1.379$ (78/114)	Ratio ≥ 1.50	$k \geq 1$	Failed significance threshold. Adjustments for backup quarterbacks must be handled via likelihood weighting rather than raw variance shifts.
Hawkes Momentum Process	$\hat{n} = 0$, Poisson behavior	$\hat{n} > 0$ (Self-excitation)	kill	In-game momentum exhibits no self-exciting characteristics.

Item / Signal / Parameter	Current Value / State	Target / Kill Threshold	Operational Tag	Rationale and Disposition
				Model remains strictly Poisson with kickoff inhibition.
Weather: Passing Wind Effect	-3.007 yds/mpg	Validated physical impact	keep	Highly predictive for passing yardage distributions. Restricted to player props; excluded from totals to avoid double-counting.
Weather: Rain Incompletion	30.1% vs. 27.6% base	Realized separation	keep	Retained for micro-level completion rate modeling.
Red-Zone TD Variance	$\text{VMR} = 1.82 \text{--} 1.92$	$\text{VMR} = 1.0$ (Poisson)	keep	Confirms Negative Binomial distribution for red-zone touchdowns. Rejects Poisson assumption across all TD props.
Spread $\times$ Total Correlation	Spearman $\rho = 0.016$	Significant linear dependence	kill	Confirms empirical independence between game spreads and game totals under standard regimes.
ATS $\times$ Over Independence	Lift = 1.011 (n=6,682)	Independence rejection	keep	Demonstrates structural independence across full game outcomes.
Post-Cover Live Total Drag	Over rate = 0.538 after 14pt cover	Exploitable market mispricing	keep	Requires withholding automated live totals following large spread covers.
Features W5–W8 (Move-37)	Wasserstein, DFA, ID, Perm Ent	Pre-registered null rejection	kill	All four feature families failed empirical

Item / Signal / Parameter	Current Value / State	Target / Kill Threshold	Operational Tag	Rationale and Disposition
				validation across 12/12 test fixtures. Fully terminated.
Inverted UX Confidence	Brier = 0.3617, z = -10.7	Brier $\leq$ 0.204	kill	Severely anti-predictive. Unpublish immediately from all customer surfaces.
Model-Ordering Switch	KL3 Brier: Market 0.2189 < Model 0.2462	Model beats Market	founder-decide	Model sorting underperforms devigged market fair probabilities. Requires founder decision to enforce marketFairProb sort.
CLV Denominator Definition	23.2% (All) vs. 40.8% (Decided)	Floor $\geq$ 52.4%	f[span_132](start_span)[span_132](end_span)ounder-decide	Requires founder sign-off to standardize reporting definitions and publish both metrics openly.
MLB Run-Line Consensus	consensusPct = 1.0	Non-tautological distribution	kill	Artifact of automated single-source pipeline. Purge from UI and database.
Serverless Odds Query	1.64M rows fetched (OOM crash)	Heap < 512\text{MB}	kill	Uncapped ingestion query caused total system failure. Replace via PR #868 batching.

Jev arXiv Research Mill Ingestion

| arXiv ID | Method / Title | Category | Operational Tag | Concrete Impact on Platform | | :--- | :--- | :--- | :--- | :--- || **1905.02928** | Predictive Inference with the Jackknife+ | UQ Conformal | keep | Deploys out-of-fold ensemble predictive intervals without single-split data burning on $n \approx 150$ windows. || **2503.01495** | Improving Efficiency of Cross-Conformal | UQ Conformal | keep | Provides bounded coverage guarantees, allowing cross-conformal sets to replace split conformal across production boards. || **1707.03307** | Fast Calibrated Additive Quantile Regression | Props / Tails | keep | Powers quantile regression GAM (qgam) for non-linear player

prop distributions and extreme tail estimation. | | **1710.02824** | Beating the Bookies with Their Own Numbers | Microstructure | keep | Implements devigging mechanics to extract clean consensus fair values across competing sportsbook feeds. | | **1211.4000** | Performance of Betting Lines in the NFL | Microstructure | keep | Establishes the historical baseline for home underdog line performance and closing line price efficiency. | | **2109.07581** | Impact of COVID-19 on Betting Markets | Microstructure | measure | Benchmarks market efficiency collapses during crowd-free regimes. Retained for historical study; not a production model. | | **2306.01740** | Correction Study of Online Sportsbooks | Microstructure | keep | Implements corrections for selection-effect bias that artificially inflates reported historical returns. | | **2607.14430** | Prices, Probabilities, and Parlays | Props / SGPs | keep | Quantifies market overpricing on multi-leg parlays relative to leg products, anchoring pricing on time-to-expiry. | | **1904.06019** | Conformal Prediction Under Covariate Shift | UQ Online | keep | Employs density-ratio likelihood weighting to rescale conformal intervals when starting quarterbacks change. | | **2202.13415** | Conformal Prediction Beyond Exchangeability | UQ Online | keep | Replaces i.i.d. exchangeability assumptions with non-exchangeable weighted quantiles over ordered weekly schedules. | | **2203.15885** | Split Conformal on Non-Exchangeable Data | UQ Online | keep | Implements sequential split-conformal methods designed for serially dependent sports time series. | | **2212.03463** | Sequential Predictive Conformal (SPCI) | UQ Online | keep | Exploits residual autocorrelation across weekly game results to adaptively tighten prediction bands. | | **2106.00170** | Adaptive Conformal Inference (ACI) | UQ Online | keep | Provides online α -adjustment to ensure rolling coverage targets hold across seasonal shifts. | | **2208.02814** | Conformal Risk Control | UQ Risk | keep | Enforces monotone risk control over expected CLV drawdowns and capital allocation thresholds. | | **2306.02732** | Conformal Prediction with Missing Values | UQ Robust | keep | Maintains distribution-free coverage when injury, roster, or weather features are missing. | | **1905.03222** | Conformalized Quantile Regression (CQR) | UQ Conformal | measure | Evaluates asymmetric prediction intervals against standard Gaussian spread margins. | | **1211.0025** | Venn-Abers Predictors | Calibration | keep | Computes distribution-free calibrated probability intervals for moneyline outcomes and binary event cards. | | **2608.11505** | Structural Models vs. Closing Price | Engine | measure | Benchmarks whether physical structural models add informational value beyond closing lines. | | **1002.0797** | Soccer: Is Scoring a Poisson Process? | Props | measure | Validates rejection of Poisson processes in sports scoring tails. Supports Negative Binomial modeling. | | **1807.07536** | Skellam Regression for Sports Margins | Margins | measure | Evaluates discrete bivariate margin generation. Soccer parameters barred from direct NFL use. | | **2604.01491** | Opponent-Adjusted Line Play in NFL | Trenches | measure | Evaluates pass blocking and pass rushing metrics to capture pressure differentials. | | **2003.02743** | Generalized Kelly Under Temporal Correlation | Staking / Ops | measure | Adjusts capital allocation models to account for covariance across correlated weekend game slates. | | **2410.09180** | Convergence & Stationarity of Elo Systems | Engine | measure | Formulates theoretical bounds on weekly rating volatility and convergence rates. | | **1810.03383** | Illusion of Persistence in Sports Data | Engine | measure | Confirms that win streaks reflect variance rather than skill, validating mean-reverting luck filters. | | **0706.3188** | Tutorial on Conformal Prediction | Reference | keep | Serves as foundational reference documentation for platform uncertainty quantification. | | **2609.10737** | Conformal Calibration Transfer (TCC) | UQ Transfer | measure | Evaluates transport-based calibration adjustments for shifts between regular season and playoff regimes. | | **2609.11935** | Physics-Informed Conformal Prediction | UQ Physical | measure | Informs boundary condition modeling for physical game-state simulations. | | **2609.12386** | Split Conformal with Label-Shift Adjustments | Calibration | measure | Provides analytical adjustments for Bayesian scores under severe

population label shifts. || **2609.13453** | Machine Learning API for Data Cubes | Architecture | kill | Geospatial earth observation API. Irrelevant to sports prediction pipelines. || **NCAAF Corpus** | College Football Betting & PBP Archive | Data Source | buy | arXiv contains no actionable NCAAF calibration papers. Acquire commercial access or scrape cfbfastR. |

F. Immediate Founder Unblocks

Engineering execution is blocked pending executive authorization on code merges, script executions, database mutations, and external data licensing. The following operational directives require immediate founder action.

Next 12 Hours: Immediate Merges and Database Remediation

1. Pull Request Merge Order

Execute in strict sequence on repository Beexly/Sports:

- **Merge PR #868 (Ops/OOM Fix):** Introduces cursor-based chunking (take: 5000, max: 50000) on odds_line_snapshots ingest queries and raises serverless function memory allocation to 1024MB. *Pre-condition:* Confirm that CI run #2948 is green.
- **Merge PR #867 (EPA Path & Decisions Register):** Enforces NFL_EPA_MIN_GAMES=4 and installs the permanent decision registry at docs/ops/FOUNDER_DECISIONS.md.
- **Merge PR #866 (Archive Freshness Monitoring):** Deploys monitoring cron alerts for data staleness exceeding 120 minutes.

2. Administrative API Executions

Execute the following API calls from the administrative deployment environment using production authorization tokens:

```
# Verify ingestion route operates without memory overflow
curl -X GET
"https://api.galaxysportsedge.com/api/cron/odds-loader?dryRun=1&limit=50" \
  -H "Authorization: Bearer ${ADMIN_SERVICE_KEY}"

# Execute [span_141] (start_span) [span_141] (end_span) dry-run backfill
of uncalibrated true probabilities (sample check)
curl -X GET
"https://api.galaxysportsedge.com/api/cron/backfill-independent-trueprob?force=1&dryRun=1&limit=5" \
  -H "Authorization: Bearer
${ADMIN_SER[span_142] (start_span) [span_142] (end_span) VICE_KEY}"

# Execute production database backfill across historical records
curl -X GET
"https://api.galaxysportsedge.com/api/[span_143] (start_span) [span_143]
(end_span) cron/backfill-independent-trueprob?force=1&dryRun=0&limit=500" \
```

```
-H "Authorization: Bearer ${ADMIN_SERVICE_KEY}"
```

3. Database Diagnostic Queries (Read-Replica)

Execute the following read queries against the PostgreSQL read-replica to quantify historical data integrity issues:

```
-- Audit the August 23 to September 12 archive gap
SELECT
    DATE(recorded_at) AS snapshot_day,
    sport,
    COUNT(*) AS total_snapshots,
    COUNT(DISTINCT event_id) AS distinct_events
FROM odds_line_snapshots
WHERE recorded_at >= '2026-08-20' AND recorded_at <= '2026-09-15'
GROUP BY DATE(recorded_at), sport
ORDER BY snapshot_day ASC;
```

```
-- Identify corrupted fixture triplications
SELECT external_id, sport, COUNT(*) as count
FROM games
WHERE scheduled_at >= '2026-08-01'
GROUP BY external_id, sport
HAVING COUNT(*) > 1;
```

```
-- Locate unfinalized January fixtures
SELECT id, sport, scheduled_at, status, updated_at
FROM games
WHERE status = 'PENDING' AND scheduled_at < '2026-02-01';
```

```
-- Identify non-unique, identical trueProb records
SELECT game_id, bet_type, true_prob, COUNT(*)
FROM pick_predictions
WHERE created_at >= '2026-09-01'
GROUP BY game_id, bet_type, true_prob
HAVING COUNT(*) > 1;
```

4. UI Display Overrides

Deploy the following feature flag overrides directly to environment configurations:

```
{
  "DISPLAY_INVERTED_CONFIDENCE": false,
  "CARD_SORT_KEY": "marketFairProb",
  "MLB_RUNLINE_CONSENSUS_SUPPRESS": true,
  "MLB_SPREAD_DEPTH_WEIGHT": 0.0,
  "PASS_RECOMMENDATIONS_VISIBLE": false,
  "NCAAF_PUBLIC_BOARD_ENABLED": false
}
```

}

Next 14 Days: Production Contracts and Data Licensing

1. Production Model Constants Sign-Off (Engine v5.3.0)

The founder must formally sign off on the production configuration parameters in packages/prediction-engine/src/config.ts prior to the scheduled October 8 auto-activation:

```
export const ENGINE_CONSTANTS_V530 = {  
  MODEL_VERSION: "5.3.0-prod",  
  NFL_SPREAD_SCALE: 45.42,           // Empirically verified modern  
scoring fold (NOT pooled 36.61)  
  NFL_HOME_FIELD_POINTS: 2.10,       // Modern HFA parameter (Replaces  
repository default 0.025)  
  DAVE_SEASON_WEIGHT: 8,             // Production prior weight k=8  
(Fay-Herriot k=7.486)  
  EPA_MIN_GAMES: 4,                 // Hard floor: no unadjusted EPA  
before Week 5  
  BOOKMAKER_CONSENSUS_WEIGHT: 0.00, // Zero out uncalibrated depth  
score  
  CONFORMAL_ALPHA: 0.10,            // 90% marginal coverage target  
  DEVIG_METHOD: "SHIN",             // Standardized market fair value  
extraction  
};
```

2. CLV Operational Policy Sign-Off

Approve the following standardized reporting contract for public-facing closing line value:

$\text{CLV}_{\text{Decided}} = \frac{\sum \mathbf{1}_{\text{Matched Beats Close}}}{\text{Total Decided Bets with Verified Close}} = 40.8\%$
 $\text{CLV}_{\text{System}} = \frac{\sum \mathbf{1}_{\text{Matched Beats Close}}}{\text{Total Published Recommendations}} = 23.2\%$

The public dashboard must display both figures alongside clear explanatory documentation.

Modifying calculation logic to artificially inflate CLV is strictly prohibited.

3. Data Vendor Acquisition Matrix

To eliminate data bottlenecks across active sports, acquire the following data licenses:

Sport / Domain	Vendor / Source	Granularity	License / Access Type	Technical Application
NCAAF Historical	CollegeFootballData (CFBD)	Play-by-play, drive metrics, historical lines (2014–2025)	Commercial Tier (\$50/mo) / Open API	Powers historical model training and baseline validation.
NCAAF Live Data	cfbfastR Pipeline	Live box scores, roster updates,	Open Source (MIT License)	Provides live ingestion features

Sport / Domain	Vendor / Source	Granularity	License / Access Type	Technical Application
		drive-by-drive metrics		during active college slates.
NFL Player Props	Sportradar / The Odds API	Real-time multi-book prop quotes (receptions, passing yards, TDs)	Commercial Production Tier (Monthly SaaS)	Feeds Negative Binomial distribution engines with live book lines.
NFL Line Movement	Pinnacle / Circa Direct Feeds	Real-time line movements with millisecond-stamped updates	API Access Agreement / B2B Commercial	Establishes ground-truth closing benchmarks for CLV calculations.
Advanced Trenches	ESPN Analytics via nflverse	Pass Block Win Rate (PBWR) & Pass Rush Win Rate (PRWR)	Open Data Commons (CC-BY 4.0)	Drives opponent-adjusted trench play modules.

G. Anti-Plan: Negative Engineering Constraints

Executing any of the following anti-patterns will compromise platform calibration and degrade overall system performance. These constraints are enforced as continuous-integration failure conditions across the repository:

- **Do Not Assign Point Margin Mean (lmu) to DAVE:** DAVE ($k=8$) achieves a CRPS of 7.500, underperforming closing market prices (7.109). DAVE must remain an auxiliary Rung-2 input for early-season adjustment, not a primary mean estimator.
- **Do Not Linearly Average Historical and Modern Scale Factors:** Averaging the historical corpus OLS scale (36.61) with the modern Hermes fold (45.42) produces a compromised value of 41.01 that miscalibrates probability density across all point spread margins.
- **Do Not Retune $k=4$ Based on Single-Fold Validation:** The 2024 fold suggested $k=4.75$, but historical walk-forward cross-validation ($n = 1,871$) confirms that $k=8$ provides superior long-term performance across full market cycles. Retuning to $k=4$ constitutes sample overfitting.
- **Do Not Flip Sorting Controls to Raw Model Output:** In competitive trials ($n = 689$), sorting by model probability yielded an inferior Brier score (0.2462) relative to devigged market consensus (0.2189). Sorting must remain anchored to market fair values until model predictions demonstrably beat the close out of sample.
- **Do Not Disguise Inverted Confidence as an "Edge Index":** Relabeling an anti-predictive metric ($z = -10.7$) while continuing to format it as a percentage misleadingly communicates certainty to customers. Any uncalibrated index must be displayed as a non-probabilistic integer score.
- **Do Not Transfer Association Football Models to American Football:** Do not adapt Dixon-Coles bivariate Poisson models, zero-inflation parameters, or goal-timing processes to NFL or college football scoring. American football scoring exhibits discrete, high-variance boundary conditions that Poisson processes fail to model.

- **Do Not Reopen Hawkes Processes as Momentum Signals:** The Hawkes process investigation demonstrated a self-excitation parameter of $\hat{\alpha} = 0$, rejecting the presence of measurable in-game momentum.
- **Do Not Reintroduce Features W5–W8:** Wasserstein distance, Detrended Fluctuation Analysis, Intrinsic Dimension, and Permutation Entropy failed pre-registered empirical trials across 12 out of 12 test fixtures. They are permanently removed from feature stores.
- **Do Not Deploy Venn-Abers Corrections Over Inverted Metrics:** Venn-Abers calibration guarantees valid probability intervals over calibrated nonconformity scores, but it cannot fix fundamentally misranked, anti-predictive scoring metrics.
- **Do Not Deploy Adaptive Conformal Inference if Window Widths Fluctuate Widely:** If early-season uncertainty intervals expand by more than 50% relative to stable historical distributions, disable adaptive conformal inference and fall back to static historical calibration quantiles.
- **Do Not Quote Narrative Features Without Empirical Measurement:** Features such as unverified player equipment adjustments, generalized fourth-down heuristics, or untracked training camp narratives must not be incorporated into pricing engines.
- **Do Not Bypass the Early-Season EPA Data Floor:** Enforce `NFL_EPA_MIN_GAMES=4` strictly. Computing opponent-adjusted EPA from small early-season samples introduces substantial estimation error into team ratings.
- **Do Not Expand Model Scope into New Sports Prematurely:** Prohibit engineering work on NBA, NHL, or soccer models until NFL, NCAAF, and MLB pipelines demonstrate full calibration and operational stability.
- **Do Not Market Platform Features as Artificial Intelligence:** Prohibit the use of vague marketing terms like "AI" or "deep neural network edge" in documentation, UI copy, and marketing materials. GSE is a sports analytics platform grounded in probability theory and empirical physics.
- **Do Not Treat Early-Season Game Samples as Season-Long Indicators:** The 17 games completed in Weeks 1 and 2 of 2026 provide an insufficient sample for updating macro-level parameters. Prior parameters must remain anchored to the historical 14,251 fixture corpus.
- **Do Not Use Small Live Settlement Samples to Claim Statistical Calibration:** Seventy live settled games cannot establish statistical calibration ($n \geq 1,000$ required). Historical backtest records must be transparently incorporated into public calibration reporting.

H. The Recomputable Standard: What "100" Looks Like

The Formal Audit Statement

When Galaxy Sports Edge achieves a readiness score of 100, its public methodology statement will be verifiable by third-party auditors and quantitative market participants:

"Galaxy Sports Edge provides real-time, distribution-free sports forecasting calibrated across 14,251 decided NFL fixtures (1999–2025) and verified against live market prices. Over $n \geq 1,000$ production selections, its observed outcome frequencies match forecast true probabilities with an Expected Calibration Error below 0.025 and a Brier score superior to closing sportsbook consensus (< 0.204). Its marginal 90% prediction intervals maintain verified coverage of 0.90

\pm 0.015 via non-exchangeable conformal inference. Across all published recommendations, it documents a positive Closing Line Value beat-rate exceeding 52.4% under immutable, timestamped accounting. All underlying datasets, execution logs, and validation code are publicly executable via open cryptographic receipts without proprietary dependencies."

Public Independent Audit Pipeline

The platform enforces public verifiability through an open four-stage verification architecture that decouples mathematical validation from internal application state.

1. Cryptographic Ledger (/performance/audit)

Every published pick generates an immutable, append-only record secured via SHA-256 hash chaining:

$$\text{Receipt}_t = \text{SHA256}(\text{PickID}_t \parallel \text{Timestamp}_t \parallel \text{GameID}_t \parallel \text{OddsSnap}_{187}(\text{start_span})(\text{end_span})\text{shot}_t \parallel \text{ModelProb}_t \parallel \text{Receipt}_{t-1})$$

Timestamps are certified through an external RFC-3161 compliant timestamping authority to establish unalterable proof of publication prior to event start times.

2. Standardized Audit Inputs

Public audit records provide clean, tabular event vectors:

- **Selection Vector:** Timestamp, Sport, GameID, BetType, Selection, PublishedLine, ModelProb, ConformalLower, ConformalUpper.
- **Closing Market Vector:** Final consensus snapshot captured at kickoff (Pinnacle and Circa midpoints), VigFreeFairProb, FinalMarketLine.
- **Realized Outcome Vector:** Official league-settled final scores, box scores, player statistics, and settlement timestamps.

3. Automated Recomputation Engine

Auditors verify platform metrics by executing standardized mathematical calculations:

$$\begin{aligned} \text{ECE} &= \sum_{m=1}^M \frac{\|B_m\|}{N} \left\| \text{acc}(B_m) - \text{conf}(B_m) \right\| \\ \text{Brier Score} &= \frac{1}{N} \sum_{t=1}^N (P_t - Y_t)^2 \\ \text{Conformal Empirical Coverage} &= \frac{1}{N} \sum_{t=1}^N \mathbf{1}_{Y_t \in [L(X_t), U(X_t)]} \\ \text{CLV}_{\text{Decided}} &= \frac{\sum_{t=1}^N \mathbf{1}_{\text{decided}}}{N_{\text{decided}}} \\ \text{CLV}_{\text{System}} &= \frac{\sum_{t=1}^N \mathbf{1}_{\text{Beats Close}}}{N_{\text{total}}} \\ \text{CLV}_{\text{System}} &= \frac{\sum_{t=1}^N \mathbf{1}_{\text{Beats Close}}}{N_{\text{total}}} \end{aligned}$$

4. Audit Execution Script

A self-contained Python script is provided to enable public verification without internal dependencies:

```
import numpy as np
import pandas as pd

def run_public_audit(export_url: str):
```

```

df = pd.read_csv(export_url)

# 1. Brier Score Audit
brier = np.mean((df["model_prob"] - df["realized_outcome"]) ** 2)

# 2. Expected Calibration Error (ECE) Audit across 10 deciles
df["bin"] = pd.qcut(df["model_prob"], q=10, duplicates="drop")
grouped = df.groupby("bin", observed=False).agg(
    conf=("model_prob", "mean"),
    acc=("realized_outcome", "mean"),
    count=("realized_outcome", "count")
)
ece = np.sum((grouped["count"] / len(df)) * np.abs(grouped["acc"]
- grouped["conf"]))

# 3. Conformal Coverage Audit
covered = (df["realized_outcome"] >= df["conformal_lower"]) &
(df["realized_outcome"] <= df["conformal_upper"])
coverage = np.mean(covered.astype(float))

# 4. Dual CLV Accounting Audit
decided = df[df["matched_close_available"] == True]
clv_decided = np.mean(decided["beat_closing_price"].astype(float))
clv_system = np.sum(df["beat_closing_price"].astype(float)) /
len(df)

return {
    "brier_score": brier,
    "ece": ece,
    "coverage_90": coverage,
    "clv_decided": clv_decided,
    "clv_system": clv_system
}

if __name__ == "__main__":
    results =
run_public_audit("https://api.galaxysportsedge.com/api/v1/audit/export
?format=csv")
    for metric, value in results.items():
        print(f"{metric}: {value:.4f}")

```

In-Scope vs. Out-of-Scope Production Boundaries

To maintain rigorous calibration standards, platform scope is strictly partitioned between certified production offerings and uncertified markets:

- **In-Scope Production Markets (Phase 4):**
 - *NFL Mainlines:* Pre-game point spreads, moneylines, and totals.

- *NFL Player Props*: Passing touchdowns, receptions, and passing yards parameterized via Negative Binomial distributions.
- *MLB Mainlines*: Pre-game moneylines and totals.
- *NCAAF Mainlines*: Division I FBS point spreads and totals (restricted to Week 5 onwards, conditional on verified data pipelines).
- **Out-of-Scope Markets (Explicitly Prohibited)**:
 - *In-Play / Live Betting*: Live betting markets across all sports.
 - *Same-Game Parlays (SGPs)*: Multi-leg correlated parlays marketed as positive-expectation bets.
 - *Professional Basketball & Hockey*: NBA, WNBA, and NHL markets (prohibited until primary sports meet institutional standards).
 - *Association Football*: MLS, EPL, and European soccer leagues.
 - *Lower-Division Football*: FCS football, Division II/III, and spring developmental leagues.
 - *Unquantified Subjective Props*: Anytime touchdown scorers, defensive player props, and novelty event markets.

Failure Thresholds for Institutional Certification

The public auditing interface enforces binary pass/fail standards. Exceeding any of the following error thresholds results in immediate suspension of platform certification badges:

- **Brier Score Degradation**: If moneyline predictions yield a Brier score exceeding 0.204 over $n \geq 1,000$ decisions, the platform's "Market-Beating" status is revoked.
- **Calibration Error Expansion**: If Expected Calibration Error exceeds 0.025 across decile bins, pick generation is automatically disabled.
- **Conformal Coverage Breach**: If marginal coverage over rolling 500-pick windows drops below 88.5% or exceeds 91.5% at nominal $\alpha = 0.10$, prediction bands are flagged as uncalibrated.
- **Cryptographic Mismatch**: If independent recomputation of receipt hashes detects discrepancies with published ledger states, platform reporting is frozen pending security and data audit.

Works cited

1. Sports/hermes/opp-adj-epa-20260919 has new results, <https://mail.google.com/mail/?extsrc=sync&client=h&plid=ACUX6DMniNevipMOsCdJqDeIUgIIHsywzaGKjd0&mid=1a0ba002bb74cacc> 2. [Beexly/Sports] Model-accuracy leaderboard — rank our own model versions by calibrated accuracy (founder-gated) (PR #96), https://mail.google.com/mail/?extsrc=sync&client=h&plid=ACUX6DMYEUKZVORY8_6wRnV4YxPh6vCSXJjDh5k&mid=19f57562a43a8f1a 3. Handoff: C:\Users\Garrett\Sports\hermes\plans\2026-09-12_095304-gse-launch-execution-handoff.md Findings remain documented in: C:\Users\Garrett\Sports\AGENTS.md The handoff contains ..., <https://mail.google.com/mail/?extsrc=sync&client=h&plid=ACUX6DN8dHt8DY9sRBRsx3pbEyz-5MSUpI51ywA&mid=1a096223ca46fdb9> 4. [Beexly/Sports] docs(ops): master research & build charter (10,000-foot, all domains) (PR #445), <https://mail.google.com/mail/?extsrc=sync&client=h&plid=ACUX6DOi1AYoiijQK2aJ4ed5NvrabkJK-osTbsA&mid=1a0262c3109b453f> 5. DECISIONS.md, <https://drive.google.com/open?id=1EPriWStTeOSnDMTtUc-CuG07c66Cgjlr>